

T3 BSM UV terms grouped by field content

EFT order: 5; loop order: 1. Columns are determined only from BSM UV terms. Empty cells denote no UV term with that field combination.

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	S_1	S_2	$F D$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{1}{2} N_{r_1}^T C N_{r_1} M_F$	$-\text{NewScalar1}_{21} \text{NewScalar1}_{12} M_F^2$ $+\frac{1}{2} \text{Im bds1Inv1NewScalar1}_{22} \text{NewScalar1}_{12} \text{Tr bds1dCGInv1}_{21,22,22,22} \text{No wfs ralar1}_{22}^{\dagger} \text{No wfs ralar1}_{22}^{\dagger}$	$-M_F^2 \text{No wfs ralar1}_{21} \text{NewScalar1}_{21}^{\dagger}$ $+\frac{1}{2} \text{Im bdsS1Inv1NewScalar1}_{22} \text{NewScalar1}_{22} \text{Tr bds1dCGInv1}_{21,22,22,22} \text{No wfs ralar1}_{22}^{\dagger} \text{No wfs ralar1}_{22}^{\dagger}$	$+\frac{1}{2} i N_{r_1}^T C \gamma^{\mu 1} D_{\mu 1} (N_{r_1})$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{1}{2} N_{j_1, r_1}^T C N_{j_1, r_1} M_F$	$-\text{NewScalar1}_{21} \text{NewScalar1}_{12} M_F^2$ $+\frac{1}{2} \text{Im bds1Inv1NewScalar1}_{22} \text{NewScalar1}_{12} \text{Tr bds1dCGInv1}_{21,22,22,22} \text{No wfs ralar1}_{22}^{\dagger} \text{No wfs ralar1}_{22}^{\dagger}$	$-M_F^2 \text{No wfs ralar1}_{21} \text{NewScalar1}_{21}^{\dagger}$ $+\frac{1}{2} \text{Im bdsS1Inv1NewScalar1}_{22} \text{NewScalar1}_{22} \text{Tr bds1dCGInv1}_{21,22,22,22} \text{No wfs ralar1}_{22}^{\dagger} \text{No wfs ralar1}_{22}^{\dagger}$	$+\frac{1}{2} i N_{j_1, r_1}^T C \gamma^{\mu 1} D_{\mu 1} (N_{j_1, r_1})$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\overline{N_{j_1, r_1}} N_{j_1, r_1} M_F$	$-\text{NewScalar1NewScalar1}^{\dagger} M_{S_1}^2$ $+\frac{1}{2} \text{lambdaS1Inv1NewScalar1}^2 \left(\text{NewScalar1}^{\dagger} \right)^2$	$-M_F^2 \text{No wfs ralar1}_{21} \text{NewScalar1}_{21}^{\dagger}$ $+\frac{1}{2} \text{Im bdsS1Inv1NewScalar1}_{22} \text{NewScalar1}_{22} \text{Tr bds1dCGInv1}_{21,22,22,22} \text{No wfs ralar1}_{22}^{\dagger} \text{No wfs ralar1}_{22}^{\dagger}$ $+\frac{1}{2} \text{Im bdsS1Inv1NewScalar1}_{22} \text{NewScalar1}_{22} \text{Tr bds1dCGInv1}_{21,22,22,22} \text{No wfs ralar1}_{22}^{\dagger} \text{No wfs ralar1}_{22}^{\dagger}$	$+\overline{N_{j_1, r_1}} \gamma^{\mu 1} D_{\mu 1} (N_{j_1, r_1})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\overline{N_{j_1, r_1}} N_{j_1, r_1} M_F$	$-\text{NewScalar1}_{21} \text{NewScalar1}_{12} M_F^2$ $+\frac{1}{2} \text{Im bds1Inv1NewScalar1}_{22} \text{NewScalar1}_{12} \text{Tr bds1dCGInv1}_{21,22,22,22} \text{No wfs ralar1}_{22}^{\dagger} \text{No wfs ralar1}_{22}^{\dagger}$ $+\frac{1}{2} \text{Im bdsS1Inv1NewScalar1}_{22} \text{NewScalar1}_{22} \text{Tr bds1dCGInv1}_{21,22,22,22} \text{No wfs ralar1}_{22}^{\dagger} \text{No wfs ralar1}_{22}^{\dagger}$	$+\frac{1}{2} \text{lambdaS2Inv1NewScalar2}^2 \left(\text{NewScalar2}^{\dagger} \right)^2$ $-\text{NewScalar2} M_{S_2}^2 \text{NewScalar2}^{\dagger}$	$+\overline{N_{j_1, r_1}} \gamma^{\mu 1} D_{\mu 1} (N_{j_1, r_1})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\overline{N_{j_1, r_1}} N_{j_1, r_1} M_F$	$-\text{NewScalar1}_{21} \text{NewScalar1}_{12} M_F^2$ $+\frac{1}{2} \text{Im bds1Inv1NewScalar1}_{22} \text{NewScalar1}_{12} \text{Tr bds1dCGInv1}_{21,22,22,22} \text{No wfs ralar1}_{22}^{\dagger} \text{No wfs ralar1}_{22}^{\dagger}$ $+\frac{1}{2} \text{Im bdsS1Inv1NewScalar1}_{22} \text{NewScalar1}_{22} \text{Tr bds1dCGInv1}_{21,22,22,22} \text{No wfs ralar1}_{22}^{\dagger} \text{No wfs ralar1}_{22}^{\dagger}$	$-M_F^2 \text{No wfs ralar1}_{21} \text{NewScalar1}_{21}^{\dagger}$ $+\frac{1}{2} \text{Im bdsS1Inv1NewScalar1}_{22} \text{NewScalar1}_{22} \text{Tr bds1dCGInv1}_{21,22,22,22} \text{No wfs ralar1}_{22}^{\dagger} \text{No wfs ralar1}_{22}^{\dagger}$ $+\frac{1}{2} \text{Im bdsS2Inv1NewScalar2}_{22} \text{NewScalar2}_{22} \text{Tr bds1dCGInv1}_{21,22,22,22} \text{No wfs ralar1}_{22}^{\dagger} \text{No wfs ralar1}_{22}^{\dagger}$	$+\overline{N_{j_1, r_1}} \gamma^{\mu 1} D_{\mu 1} (N_{j_1, r_1})$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	D	S_1	H	S_1	S_2	S_2	D
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1}$	$(\text{NewScalar1}_{j_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{j_1})$	$+\text{lambdaH1Inv1}H_{a2}\text{NewScalar1}_{j_2}\text{T3HiggsPortal1CGInv1}_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger\text{NewScalar1}_{j_1}^\dagger$ $+\text{lambdaH1Inv2}H_{a2}\text{NewScalar1}_{j_2}\text{T3HiggsPortal1CGInv2}_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger\text{NewScalar1}_{j_1}^\dagger$	$+\text{lambdaH1Inv1NewScalar1}_{j_2}\text{NewScalar1}_{j_1}\text{T1CrossScalarCGInv1}_{j_1,j_2,j_2,a_2}\text{NewScalar1}_{j_2}^\dagger\text{NewScalar1}_{j_2}^\dagger$ $+\text{lambdaH1Inv2NewScalar1}_{j_2}\text{NewScalar1}_{j_1}\text{T1CrossScalarCGInv1}_{j_1,j_2,j_2,a_2}\text{NewScalar1}_{j_2}^\dagger\text{NewScalar1}_{j_2}^\dagger$	$+D_{\mu_1} (\text{NewScalar2}_{j_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{j_1})$			
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+D_{\mu_1}$	$(\text{NewScalar1}_{j_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{j_1})$	$+\text{lambdaH1Inv1}H_{a2}\text{NewScalar1}_{j_2}\text{T3HiggsPortal1CGInv1}_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger\text{NewScalar1}_{j_1}^\dagger$ $+\text{lambdaH1Inv2}H_{a2}\text{NewScalar1}_{j_2}\text{T3HiggsPortal1CGInv2}_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger\text{NewScalar1}_{j_1}^\dagger$	$+\text{lambdaH1Inv1NewScalar1}_{j_2}\text{NewScalar1}_{j_1}\text{T1CrossScalarCGInv1}_{j_1,j_2,j_2,a_2}\text{NewScalar1}_{j_2}^\dagger\text{NewScalar1}_{j_2}^\dagger$ $+\text{lambdaH1Inv2NewScalar1}_{j_2}\text{NewScalar1}_{j_1}\text{T1CrossScalarCGInv1}_{j_1,j_2,j_2,a_2}\text{NewScalar1}_{j_2}^\dagger\text{NewScalar1}_{j_2}^\dagger$	$+D_{\mu_1} (\text{NewScalar2}_{j_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{j_1})$			
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}$	$(\text{NewScalar1})^\dagger D_{\mu_1} (\text{NewScalar1})$	$+\text{lambdaH1Inv1NewScalar1}H_{a2}\text{T3HiggsPortal1CGInv1}_{a_1,a_2}\text{NewScalar1}^\dagger H_{a_1}^\dagger$ $+\text{lambdaH1Inv2NewScalar1}H_{a2}\text{T3HiggsPortal1CGInv2}_{a_1,a_2}\text{NewScalar1}^\dagger H_{a_1}^\dagger$	$+\text{lambdaH1Inv1NewScalar1}_{j_2}\text{NewScalar1}_{j_1}\text{T1CrossScalarCGInv1}_{j_1,j_2,j_2,a_2}\text{NewScalar1}_{j_2}^\dagger\text{NewScalar1}_{j_2}^\dagger$ $+\text{lambdaH1Inv2NewScalar1}_{j_2}\text{NewScalar1}_{j_1}\text{T1CrossScalarCGInv1}_{j_1,j_2,j_2,a_2}\text{NewScalar1}_{j_2}^\dagger\text{NewScalar1}_{j_2}^\dagger$	$+D_{\mu_1} (\text{NewScalar2}_{j_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{j_1})$			
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+D_{\mu_1}$	$(\text{NewScalar1}_{j_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{j_1})$	$+\frac{1}{2}\text{lambdaH1Inv1}H_{a2}\text{NewScalar1}_{j_2}\text{T3HiggsPortal1CGInv1}_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger\text{NewScalar1}_{j_1}^\dagger$ $+\frac{1}{2}\text{lambdaH1Inv2}H_{a2}\text{NewScalar1}_{j_2}\text{T3HiggsPortal1CGInv2}_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger\text{NewScalar1}_{j_1}^\dagger$	$+\text{lambdaH1Inv1NewScalar1}_{j_2}\text{NewScalar1}_{j_1}\text{T1CrossScalarCGInv1}_{j_1,j_2,j_2,a_2}\text{NewScalar1}_{j_2}^\dagger\text{NewScalar1}_{j_2}^\dagger$ $+\frac{1}{2}\text{lambdaH1Inv2}H_{a2}\text{NewScalar1}_{j_2}\text{T3HiggsPortal1CGInv2}_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger\text{NewScalar1}_{j_1}^\dagger$	$+D_{\mu_1} (\text{NewScalar2})^\dagger D_{\mu_1} (\text{NewScalar2})$			
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}$	$(\text{NewScalar1}_{j_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{j_1})$	$+\frac{1}{2}\text{lambdaH1Inv1}H_{a2}\text{NewScalar1}_{j_2}\text{T3HiggsPortal1CGInv1}_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger\text{NewScalar1}_{j_1}^\dagger$ $+\frac{1}{2}\text{lambdaH1Inv2}H_{a2}\text{NewScalar1}_{j_2}\text{T3HiggsPortal1CGInv2}_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger\text{NewScalar1}_{j_1}^\dagger$	$+\text{lambdaH1Inv1NewScalar1}_{j_2}\text{NewScalar1}_{j_1}\text{T1CrossScalarCGInv1}_{j_1,j_2,j_2,a_2}\text{NewScalar1}_{j_2}^\dagger\text{NewScalar1}_{j_2}^\dagger$ $+\frac{1}{2}\text{lambdaH1Inv2}H_{a2}\text{NewScalar1}_{j_2}\text{T3HiggsPortal1CGInv2}_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger\text{NewScalar1}_{j_1}^\dagger$	$+D_{\mu_1} (\text{NewScalar2}_{j_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{j_1})$			

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 H$	$S_1 F \ell$	$S_1 S_2 H$	$S_2 F \ell$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+i\text{ambda}H_2\text{Inv}1H_{a_2}\text{NewScalar}2_{j_2}\text{T3HiggsPortal}2\text{CGInv}1_{a_1,a_2,i_1,j_2}H_{a_1}^{\dagger}\text{NewScalar}2_{j_1}^{\dagger}$ $+i\text{ambda}H_2\text{Inv}2H_{a_2}\text{NewScalar}2_{j_2}\text{T3HiggsPortal}2\text{CGInv}2_{a_1,a_2,i_1,j_2}H_{a_1}^{\dagger}\text{NewScalar}2_{j_1}^{\dagger}$	$+\overline{\ell_{a_1,i_1}}P_RN_{r_1}\text{NewScalar}1_{j_1}\text{T3Y1CG}_{a_1,j_1}y_{1,i_1,r_1}$ $+\overline{y_{1,i_1,r_1}}N_{r_1}^TCPL\ell_{a_1,i_1}\text{T3Y1CG}_{a_1,j_1}\text{NewScalar}1_{j_1}^{\dagger}$	$+\overline{\lambda_{\tau_3}}\text{NewScalar}2_{j_2}\text{T3MixCG}_{a_1,a_2,j_1,j_2}H_{a_1}^{\dagger}H_{a_2}^{\dagger}\text{NewScalar}1_{j_1}^{\dagger}$ $+H_{a_1}H_{a_2}\text{NewScalar}1_{j_2}\text{T3MixCG}_{a_1,a_2,j_2,j_1}\overline{\lambda_{\tau_3}}\text{NewScalar}2_{j_1}^{\dagger}$	$+\overline{y_{2,i_1,r_1}}N_{r_1}^TCPL\ell_{a_1,i_1}\text{NewScalar}2_{j_1}\text{T3Y2CG}_{a_1,j_1}$ $+\overline{\ell_{a_1,i_1}}P_RN_{r_1}\text{T3Y2CG}_{a_1,j_1}y_{2,i_1,r_1}\text{NewScalar}2_{j_1}^{\dagger}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+i\text{ambda}H_2\text{Inv}1H_{a_2}\text{NewScalar}2_{j_2}\text{T3HiggsPortal}2\text{CGInv}1_{a_1,a_2,j_1,j_2}H_{a_1}^{\dagger}\text{NewScalar}2_{j_1}^{\dagger}$ $+i\text{ambda}H_2\text{Inv}2H_{a_2}\text{NewScalar}2_{j_2}\text{T3HiggsPortal}2\text{CGInv}2_{a_1,a_2,j_1,j_2}H_{a_1}^{\dagger}\text{NewScalar}2_{j_1}^{\dagger}$	$+\overline{\ell_{a_1,i_1}}P_RN_{j_1,r_1}\text{NewScalar}1_{j_1}\text{T3Y1CG}_{a_1,j_1,j_1}y_{1,i_1,r_1}$ $+\overline{y_{1,i_1,r_1}}N_{j_1,r_1}^TCPL\ell_{a_1,i_1}\text{T3Y1CG}_{a_1,j_1,j_1}\text{NewScalar}1_{j_1}^{\dagger}$	$+\overline{\lambda_{\tau_3}}\text{NewScalar}2_{j_2}\text{T3MixCG}_{a_1,a_2,j_1,j_2}H_{a_1}^{\dagger}H_{a_2}^{\dagger}\text{NewScalar}1_{j_1}^{\dagger}$ $+H_{a_1}H_{a_2}\text{NewScalar}1_{j_2}\text{T3MixCG}_{a_1,a_2,j_2,j_1}\overline{\lambda_{\tau_3}}\text{NewScalar}2_{j_1}^{\dagger}$	$+\overline{y_{2,i_1,r_1}}N_{j_1,r_1}^TCPL\ell_{a_1,i_1}\text{NewScalar}2_{j_1}\text{T3Y2CG}_{a_1,j_1,j_1}$ $+\overline{\ell_{a_1,i_1}}P_RN_{j_1,r_1}\text{T3Y2CG}_{a_1,j_1,j_1}y_{2,i_1,r_1}\text{NewScalar}2_{j_1}^{\dagger}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{2}i\text{ambda}H_2\text{Inv}1H_{a_2}\text{NewScalar}2_{j_2}\text{T3HiggsPortal}2\text{CGInv}1_{a_1,a_2,i_1,j_2}H_{a_1}^{\dagger}\text{NewScalar}2_{j_1}^{\dagger}$ $+\frac{1}{2}i\text{ambda}H_2\text{Inv}1H_{a_2}\text{NewScalar}2_{j_2}\text{T3HiggsPortal}2\text{CGInv}1_{a_1,a_2,j_2,j_1}H_{a_1}^{\dagger}\text{NewScalar}2_{j_1}^{\dagger}$ $+\frac{1}{2}i\text{ambda}H_2\text{Inv}2H_{a_2}\text{NewScalar}2_{j_2}\text{T3HiggsPortal}2\text{CGInv}2_{a_1,a_2,i_1,j_2}H_{a_1}^{\dagger}\text{NewScalar}2_{j_1}^{\dagger}$ $+\frac{1}{2}i\text{ambda}H_2\text{Inv}2H_{a_2}\text{NewScalar}2_{j_2}\text{T3HiggsPortal}2\text{CGInv}2_{a_1,a_2,j_2,j_1}H_{a_1}^{\dagger}\text{NewScalar}2_{j_1}^{\dagger}$	$+\text{NewScalar}1\overline{\ell_{a_1,i_1}}CP_R\overline{N_{j_1,r_1}}^T\text{T3Y1CG}_{a_1,j_1}y_{1,i_1,r_1}$ $+\overline{y_{1,i_1,r_1}}\ell_{a_1,i_1}^TCPLN_{j_1,r_1}\text{T3Y1CG}_{a_1,j_1}\text{NewScalar}1^{\dagger}$	$+\overline{\lambda_{\tau_3}}\text{NewScalar}2_{j_1}\text{T3MixCG}_{a_1,a_2,j_1}\text{NewScalar}1^{\dagger}H_{a_1}^{\dagger}H_{a_2}^{\dagger}$ $+\text{NewScalar}1H_{a_1}H_{a_2}\text{T3MixCG}_{a_1,a_2,j_1}\overline{\lambda_{\tau_3}}\text{NewScalar}2_{j_1}^{\dagger}$	$+\overline{y_{2,i_1,r_1}}\overline{N_{j_1,r_1}}P_L\ell_{a_1,i_1}\text{NewScalar}2_{j_1}\text{T3Y2CG}_{a_1,j_1,j_1}$ $+\overline{\ell_{a_1,i_1}}P_RN_{j_1,r_1}\text{T3Y2CG}_{a_1,j_1,j_1}y_{2,i_1,r_1}\text{NewScalar}2_{j_1}^{\dagger}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+i\text{ambda}H_2\text{Inv}1\text{NewScalar}2H_{a_2}\text{T3HiggsPortal}2\text{CGInv}1_{a_1,a_2}\text{NewScalar}2^{\dagger}H_{a_1}^{\dagger}$	$+\overline{\ell_{a_1,i_1}}CP_R\overline{N_{j_1,r_1}}^T\text{NewScalar}1_{j_1}\text{T3Y1CG}_{a_1,j_1,j_1}y_{1,i_1,r_1}$ $+\overline{y_{1,i_1,r_1}}\ell_{a_1,i_1}^TCPLN_{j_1,r_1}\text{T3Y1CG}_{a_1,j_1,j_1}\text{NewScalar}1_{j_1}^{\dagger}$	$+H_{a_1}H_{a_2}\text{NewScalar}1_{j_1}\text{T3MixCG}_{a_1,a_2,j_1}\overline{\lambda_{\tau_3}}\text{NewScalar}2^{\dagger}$ $+\text{NewScalar}2\overline{\lambda_{\tau_3}}\text{T3MixCG}_{a_1,a_2,j_1}H_{a_1}^{\dagger}H_{a_2}^{\dagger}\text{NewScalar}1_{j_1}^{\dagger}$	$+\text{NewScalar}2\overline{y_{2,i_1,r_1}}\overline{N_{j_1,r_1}}P_L\ell_{a_1,i_1}\overline{\text{T3Y2CG}}_{a_1,j_1}$ $+\overline{\ell_{a_1,i_1}}P_RN_{j_1,r_1}\text{T3Y2CG}_{a_1,j_1}y_{2,i_1,r_1}\text{NewScalar}2^{\dagger}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{2}i\text{ambda}H_2\text{Inv}1H_{a_2}\text{NewScalar}2_{j_2}\text{T3HiggsPortal}2\text{CGInv}1_{a_1,a_2,i_1,j_2}H_{a_1}^{\dagger}\text{NewScalar}2_{j_2}^{\dagger}$ $+\frac{1}{2}i\text{ambda}H_2\text{Inv}1H_{a_2}\text{NewScalar}2_{j_2}\text{T3HiggsPortal}2\text{CGInv}1_{a_1,a_2,j_2,j_1}H_{a_1}^{\dagger}\text{NewScalar}2_{j_1}^{\dagger}$ $+\frac{1}{2}i\text{ambda}H_2\text{Inv}2H_{a_2}\text{NewScalar}2_{j_2}\text{T3HiggsPortal}2\text{CGInv}2_{a_1,a_2,i_1,j_2}H_{a_1}^{\dagger}\text{NewScalar}2_{j_1}^{\dagger}$ $+\frac{1}{2}i\text{ambda}H_2\text{Inv}2H_{a_2}\text{NewScalar}2_{j_2}\text{T3HiggsPortal}2\text{CGInv}2_{a_1,a_2,j_2,j_1}H_{a_1}^{\dagger}\text{NewScalar}2_{j_1}^{\dagger}$	$+\overline{\ell_{a_1,i_1}}CP_R\overline{N_{j_1,r_1}}^T\text{NewScalar}1_{j_1}\text{T3Y1CG}_{a_1,j_1,j_1}y_{1,i_1,r_1}$ $+\overline{y_{1,i_1,r_1}}\ell_{a_1,i_1}^TCPLN_{j_1,r_1}\text{T3Y1CG}_{a_1,j_1,j_1}\text{NewScalar}1_{j_1}^{\dagger}$	$+\overline{\lambda_{\tau_3}}\text{NewScalar}2_{j_2}\text{T3MixCG}_{a_1,a_2,j_1,j_2}H_{a_1}^{\dagger}H_{a_2}^{\dagger}\text{NewScalar}1_{j_1}^{\dagger}$ $+H_{a_1}H_{a_2}\text{NewScalar}1_{j_2}\text{T3MixCG}_{a_1,a_2,j_2,j_1}\overline{\lambda_{\tau_3}}\text{NewScalar}2_{j_1}^{\dagger}$	$+\overline{y_{2,i_1,r_1}}\overline{N_{j_1,r_1}}P_L\ell_{a_1,i_1}\text{NewScalar}2_{j_1}\text{T3Y2CG}_{a_1,j_1,j_1}$ $+\overline{\ell_{a_1,i_1}}P_RN_{j_1,r_1}\text{T3Y2CG}_{a_1,j_1,j_1}y_{2,i_1,r_1}\text{NewScalar}2_{j_1}^{\dagger}$